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FACULTY OF ENGINEERING AND INFORMATION SCIENCES

Group Project Coversheet

Attach this Cover Sheet as the front page of the assignment. Assignments are to be submitted to the relevant academic. Submit only one copy of the assignment per group to the relevant online site.

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Wine Quality Prediction Using Regression and Classification Models

1. Introduction

Wine is an important consumer product that is evaluated not only for consumption, but also for taste, aroma, balance, and perceived quality. Wine quality assessment is important for producers, sellers, and consumers because it affects product consistency, certification, pricing, and market value. Traditionally, wine quality is determined through human sensory evaluation, where tasters judge characteristics such as flavour, acidity, sweetness, aroma, mouthfeel, and overall drinking experience. However, sensory evaluation can be subjective because different tasters may perceive wine characteristics differently (Cortez et al., 2009).

To support a more objective and consistent evaluation process, machine learning provides a data-driven approach for predicting wine quality using physicochemical measurements. Instead of relying only on human judgement, machine learning models can learn patterns between chemical properties and wine quality scores from historical data. In this project, the UCI Wine Quality dataset was used, containing physicochemical measurements for Portuguese “Vinho Verde” red and white wines (Cortez et al., 2009). The red wine dataset contains 1,599 samples, while the white wine dataset contains 4,898 samples. Each wine sample includes 11 physicochemical features, such as alcohol, acidity, sulphates, pH, density, and sulfur dioxide levels, together with a sensory quality score ranging from 0 to 10.

Wine quality prediction can be approached as both a regression and classification problem. Regression models predict continuous wine quality scores, while classification models group wines into broader categories such as Low, Medium, and High quality. Several machine learning algorithms were explored in this project, including Linear Regression, Random Forest, Gradient Boosting, Logistic Regression, and Gaussian Naive Bayes.

Machine learning techniques can support wine quality control, certification, production monitoring, and decision-making by identifying relationships between physicochemical properties and perceived wine quality. The main objective of this project is to evaluate the effectiveness of different machine learning models for wine quality prediction and analyse their prediction performance, generalisation ability, and feature importance.

2. Background Theory

2.1 Identifying the Quality of Wine

Wine quality is traditionally evaluated through a sensory assessment performed by human wine tasters. Experts judge characteristics such as aroma, flavour, acidity, sweetness, balance, and aftertaste to determine the overall quality of a wine, in a blind setting and in randomized order to reduce bias. (AWRI, n.d.).

The evaluations are then averaged and converted into a quality score of scale 0–10. Physicochemical measurements are then obtained from lab testing. For this project, the UCI Wine Quality dataset (UCI Machine Learning Repository, 2009) is used which includes features such as alcohol content, acidity, pH, sulphates, density, and sulfur d

ioxide levels for identifying patterns related to wine quality, along with the respective quality score from human tasters.

Both regression and classification machine learning techniques can then be used to predict which chemical characteristics contribute most strongly to perceived wine quality to support control and production decision-making.

2.2 Data Cleaning, Preprocessing and Analysis

Before model development, both the red wine and white wine datasets were cleaned and preprocessed to improve data quality and model reliability. This process included missing and duplicate value checking, label transformation, stratified splitting, cross-validation, and feature scaling.

Missing and Duplicate Values Cleaning

Missing value checking was performed to ensure data completeness. Duplicate records were removed to reduce redundancy and minimise potential data leakage during model training.

Label Transformation (Regrouping)

Wine quality scores were regrouped into Low, Medium, and High categories to simplify the multi-class classification task and improve class distribution balance.

Stratified Split

Stratified splitting preserves the original class distribution in both training and testing datasets, improving evaluation reliability for imbalanced classification tasks.

Cross Validation

Cross-validation evaluates model stability by repeatedly training and validating the model on different data folds, providing a more reliable estimate of generalisation performance.

Feature Scaling – StandardScaler

StandardScaler was applied to Linear Regression as it is sensitive to feature scale. It standardises each feature to zero mean and unit variance:

$$z = (x - \mu) / \sigma$$

Tree-based models (Random Forest, Gradient Boosting) do not require scaling. (Scikit-learn Developers, n.d.-c)

2.3 Regression Models

Three regression models were selected to represent a range of learning approaches, from simple linear assumptions to complex ensemble methods.

Linear Regression

Linear Regression predicts the output as a weighted sum of input features, with weights optimised by minimising MSE via Ordinary Least Squares (OLS):

$$\hat{y} = w_0 + w_1 x_1 + w_2 x_2 + \dots + w_{1-1} x_{1-1}$$

$$MSE = (1/n) \sum (y_i - \hat{y}_i)^2$$

It serves as a simple baseline but may underfit complex nonlinear patterns.

Random Forest Regressor

Random Forest builds multiple independent decision trees and averages their predictions:

$$\hat{y} = (1/T) \sum f_t(x)$$

Each tree is trained via bagging, reducing overfitting compared to a single decision tree. (Breiman, 2001)

Gradient Boosting Regressor

Gradient Boosting builds trees sequentially, each correcting the residual errors of the previous:

$$F_m(x) = F_{m-1}(x) + \eta \cdot f_m(x)$$

The learning rate η controls the contribution of each tree, balancing complexity and generalisation. (Friedman, 2001)

2.4 Classification Models

Three classification models were used: Gaussian Naive Bayes, Logistic Regression, and Random Forest Classifier. They represent probabilistic, linear, and nonlinear ensemble approaches.

Gaussian Naive Bayes

GaussianNB applies Bayes' theorem to estimate the probability of each class:

$$P(C_k | x) = P(C_k)P(x | C_k) / P(x)$$

With the naive independence assumption: $P(C_k | x) \propto P(C_k) \prod P(x_j | C_k)$

The predicted class is the one with the highest posterior probability: $\hat{y} = \operatorname{argmax}_k P(C_k | x)$

GaussianNB was used as a fast probabilistic baseline, although the independence assumption may not fully hold for chemical features (Bishop, 2006; Zhang, 2004).

Logistic Regression

Logistic Regression estimates class probabilities from a linear combination of features. For multiclass classification, the softmax function is used:

$$P(y = k | x) = \exp(w_k^T x + b_k) / \sum_j \exp(w_j^T x + b_j) \quad \hat{y} = \operatorname{argmax}_k P(y = k | x)$$

It was used as a linear baseline. `RobustScaler` was applied because the model is sensitive to feature scale and outliers, and class weighting was used to reduce imbalance effects (He & Garcia, 2009).

Random Forest Classifier

Random Forest combines multiple decision trees and predicts the final class by majority voting:

$$\hat{y} = \operatorname{mode}\{h_1(x), h_2(x), \dots, h_T(x)\}$$

The class probability can also be estimated as: $P(y = k | x) = (1/T) \sum I(h_t(x) = k)$

Random Forest was used as the nonlinear ensemble model because it can capture feature interactions and reduce variance compared with a single decision tree (Breiman, 2001; Hastie et al., 2009). Since the class distribution is imbalanced, macro-F1 was used as the main metric which gives equal importance to each class, making it more s

uitable than accuracy for imbalanced wine quality categories (Powers, 2011; Saito & Rehmsmeier, 2015).

2.5 Model Evaluation Metrics and Method

Model evaluation metrics were used to measure prediction performance and generalisation ability for both regression and classification tasks. RMSE, R^2 , and residual plot analysis were applied to evaluate regression models, while Accuracy, Precision, Recall, Macro-F1 score, confusion matrix analysis, and feature importance analysis were used to evaluate classification performance and interpret the contribution of different wine features.

RMSE

Root Mean Squared Error (RMSE) measures the average prediction error between predicted and actual wine quality values:

$$RMSE = \sqrt{\frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2}$$

Lower RMSE values indicate better prediction accuracy.

R^2

R^2 measures how well the regression model explains the variation in wine quality:

$$R^2 = 1 - \frac{\sum (y_i - \hat{y}_i)^2}{\sum (y_i - \bar{y})^2}$$

Higher R^2 values indicate better model fitting performance.

Residual Plot

Residual plots visualise the prediction errors by plotting residual values against predicted values. They help identify model bias, error distribution, and potential overfitting or nonlinear patterns in the regression models.

Macro-F1 Score

Macro-F1 evaluates classification performance by averaging the F1-score of all classes equally:

$$Macro-F1 = \frac{1}{N} \sum_{i=1}^N \left(2 \times \frac{Precision_i \times Recall_i}{Precision_i + Recall_i} \right)$$

Macro averaging helps evaluate imbalanced multi-class datasets more fairly than accuracy alone. (Scikit-learn Developers, n.d.-b)

(Precision evaluates prediction correctness, while recall measures how completely the model identifies each class. F1-score balances both metrics.)

Confusion Matrix

The confusion matrix visualises the classification results by comparing the true labels with the predicted labels. It helps identify correctly classified samples and the classes that are most frequently misclassified. (Scikit-learn Developers, n.d.-a)

Feature Importances

Feature importance analysis identifies which wine characteristics contribute most to the Random Forest classification model. Higher importance scores indicate features that have a stronger influence on wine quality prediction. (Gerstorfer et al., 2023)

3. Solutions, Evaluation, and Discussions

3.1 Data Cleaning and Preprocessing

Red and white wine were analysed separately because their chemical distributions differ. For each dataset, missing values and duplicate rows were checked. Duplicate records were removed to avoid repeated samples affecting training and evaluation.

The original quality score was regrouped into three classes for classification: Low (≤ 5), Medium ($= 6$), and High (≥ 7). The original score was kept for regression. Therefore, two target variables were created: `y_cls` for classification and `y_reg` for regression.

A shared train-test split was used for both classification and regression models. This ensured that all six models were evaluated on the same training and testing samples. Stratification was applied using `y_cls` to preserve class proportions in the train and test sets.

3.2 Regression Model Setup and Fine-tune

All three regression models were trained and evaluated separately on the red and white wine datasets. An 80/20 train-test split was applied, and GridSearchCV with 5-Fold Cross-Validation was used for hyperparameter tuning, with `neg_mean_squared_error` as the scoring metric.

Linear Regression

Parameter	Search Space	Best Value (Red)	Best Value (White)
<code>fit_intercept</code>	<code>[True, False]</code>	True	True

Best CV RMSE — Red: 0.6665 / White: 0.7519

Random Forest Regressor

Parameter	Search Space	Best Value (Red)	Best Value (White)
n_estimators	[200, 400]	200	400
max_depth (Red/White)	[None, 10, 15, 20] / [None, 10, 20]	None	None
min_samples_split	[2, 5, 10]	10	2
min_samples_leaf	[1, 2, 4]	4	1
max_features	['sqrt', 0.6, 1.0]	sqrt	sqrt

Best CV RMSE — Red: 0.6400 / White: 0.6959

Gradient Boosting Regressor

Parameter	Search Space	Best Value (Red)	Best Value (White)
n_estimators	[100, 200]	100	100
learning_rate	[0.05, 0.1]	0.05	0.05
max_depth	[3, 5]	3	5

Best CV RMSE — Red: 0.6553 / White: 0.7092

3.3 Regression Model Evaluation

Performance Comparison and Overfitting Analysis

Red/White Wine	Train RMSE	Test RMSE	Train R ²	Test R ²
Linear Regression	0.6560 / 0.7443	0.6629 / 0.7505	0.3703 / 0.3003	0.3294 / 0.2947
Random Forest Regressor	0.4468 / 0.2564	0.6451 / 0.7013	0.7079 / 0.9170	0.3649 / 0.3841
Gradient Boosting Regressor	0.5388 / 0.5705	0.6441 / 0.7109	0.5752 / 0.5889	0.3670 / 0.3672

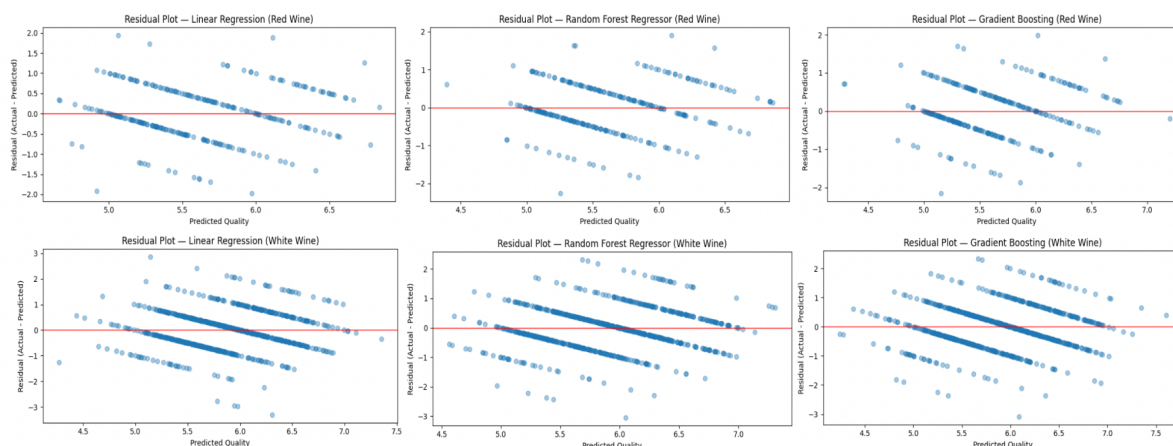
The comparison results show that all three regression models achieved relatively low Test R² values on both the red wine and white wine datasets. This suggests that the physicochemical features can only partially explain wine quality, while subjective human judgement and other hidden factors may also influence the final quality score.

Linear Regression produced the most consistent training and testing results, while Random Forest achieved very strong training performance but showed noticeable overfitting, especially in the white wine dataset where the gap between training and testing

performance was large. Gradient Boosting Regressor achieved a better balance between prediction accuracy and generalisation, with lower overfitting and the strongest overall testing performance.

Overall, the low R^2 values indicate that wine quality prediction is a complex nonlinear problem and that the current dataset may not fully capture all factors affecting wine quality.

Residual Plot



The residual plots for all three regression models show that most residuals are distributed around the zero line, indicating that the models are generally unbiased in prediction. However, several patterns can still be observed.

For both the red wine and white wine datasets, Linear Regression shows relatively consistent residual distributions, but the residual spread remains fairly large. This suggests that the linear model struggles to fully capture the nonlinear relationships between wine features and quality.

Random Forest Regressor shows tighter residual distributions around zero for many samples, especially in the training-related prediction range, indicating stronger fitting ability. However, several larger residual outliers are still visible, suggesting that the model may overfit certain patterns in the dataset.

Gradient Boosting Regressor demonstrates a more balanced residual distribution with fewer extreme residuals compared with Random Forest. The residual spread is relatively stable across prediction ranges, indicating better generalisation performance and a more balanced learning behaviour.

Across all six plots, the white wine dataset generally shows a wider residual range than the red wine dataset, suggesting that white wine quality may be more difficult to predict accurately due to higher variability and more complex feature relationships.

3.4 Classification Model Setup, Comparison and Fine-tune

The three classification models were first compared using stratified 5-fold cross-validation on the training set. Stratified cross-validation was chosen because it preserves class proportions in each fold, making the evaluation more reliable for imbalanced labels.

For both red and white wine, Random Forest achieved the highest cross-validation macro-F1 score:

Red/White Wine	CV Accuracy	CV Macro-F1
Random Forest	0.6725 / 0.6080	0.6185 / 0.5903
Logistic Regression	0.6026 / 0.5467	0.5706 / 0.5463
GaussianNB	0.6017 / 0.5022	0.5623 / 0.5070

Based on these results, Random Forest was selected for fine-tuning using GridSearchCV. The tuning objective was macro-F1 because the model needed to perform fairly across all three quality groups, not only the majority class.

For red wine, the best Random Forest parameters were max_depth=15, max_features='sqrt', min_samples_leaf=2, min_samples_split=5, and n_estimators=400. The tuned model achieved a test macro-F1 of 0.5931 and test accuracy of 0.6213. The main confusions were Low → Medium, Medium → Low, and High → Medium, showing that most errors occurred between neighbouring quality groups.

For white wine, a smaller computation-aware tuning grid was used because the dataset was larger. The final model used 500 trees with tuned structure parameters such as min_samples_leaf=4. It achieved a test macro-F1 of 0.5969 and test accuracy of 0.5990. The most common confusions were Low → Medium, Medium → Low, and High → Medium, again suggesting overlap between adjacent quality groups.

Overall, Random Forest performed best for both red and white wine classification. However, train-test gaps showed some overfitting, especially because training macro-F1 was much higher than test macro-F1. This indicates that although Random Forest captures nonlinear patterns effectively, further regularisation or additional validation strategies may improve generalisation.

3.5 Random Forest Classifier Model Evaluation

Overfitting Analysis

The following tables present the overfitting analysis results for the selected Random Forest classifier on the red wine and white wine datasets by comparing training, testing, and cross-validation macro-F1 scores. Large gaps between training and testing results indicate overfitting.

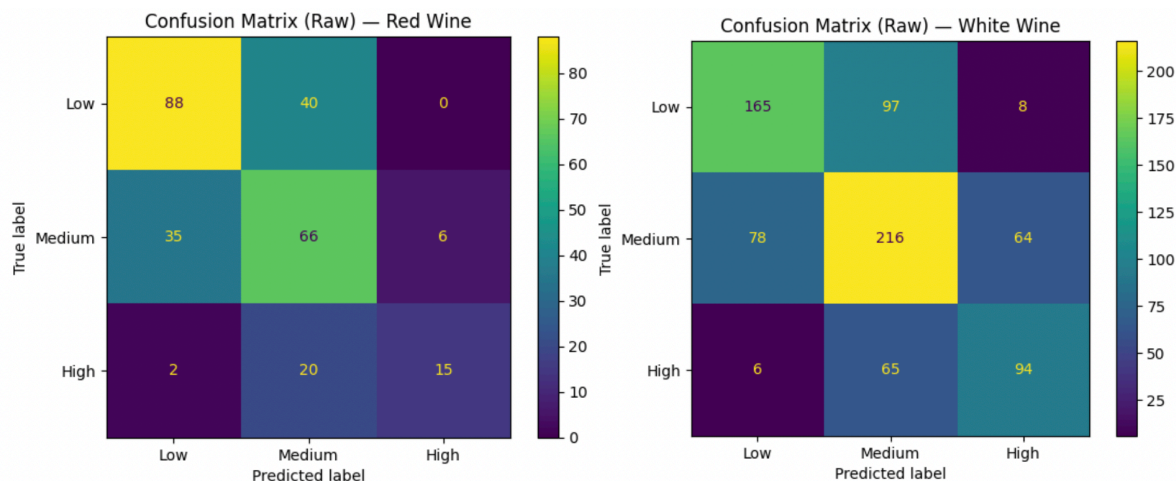
Red/White Wine	F1_Macro	Accuracy	CV macro-F1 mean
Train	0.9892 / 0.9229	0.9899 / 0.9227	0.9838 / 0.9220
Test	0.5931 / 0.5969	0.6213 / 0.5990	0.6231 / 0.5969
Gap	0.3961 / 0.3260	0.3686 / 0.3237	0.3607 / 0.3251

For both datasets, the Random Forest classifier achieved much higher training scores than testing and cross-validation scores, indicating moderate overfitting and weaker generalisation on unseen data. The red wine model showed slightly larger performance gaps than the white wine model.

Possible reasons include the relatively small dataset size and class imbalance, which may cause the model to learn dataset-specific or majority-class patterns more strongly than general relationships.

Confusion Matrix

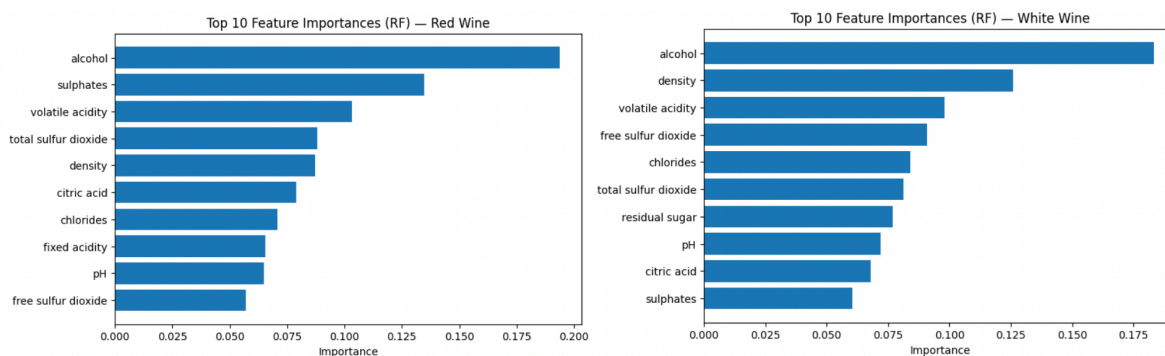
The following confusion matrices illustrate the classification performance of the selected Random Forest classifier on the red wine and white wine datasets. The matrices compare the true wine quality classes with the predicted classes, allowing clearer observation of correctly classified samples and common misclassification patterns between Low, Medium, and High wine quality categories.



Both confusion matrices show that most predictions are correctly classified along the diagonal. However, many errors occur between neighbouring classes, especially between Low and Medium or Medium and High categories. This suggests that adjacent wine quality groups share similar chemical characteristics, making them more difficult for the Random Forest classifier to distinguish clearly.

Feature Importances

The following feature importance charts show the most influential chemical attributes used by the Random Forest classifier to predict wine quality for both the red wine and white wine datasets.



For both datasets, alcohol was identified as the most important feature for predicting wine quality. In the red wine dataset, sulphates and volatile acidity also showed relatively high importance, while density was more influential in the white wine dataset. Overall, the feature importance results suggest that multiple chemical properties contribute to wine quality prediction, although the influence of each feature differs between red and white wines.

4. Conclusion

This project demonstrated that machine learning models can support wine quality prediction by learning relationships between physicochemical features and wine quality outcomes. Through comparing multiple regression and classification models, the group

p observed that stronger training performance does not always lead to better generalisation on unseen data. In particular, tree-based models such as Random Forest achieved strong fitting ability but also showed noticeable overfitting, highlighting the importance of evaluating models beyond training accuracy alone.

The project also showed that wine quality prediction is inherently challenging because neighbouring quality categories often share similar chemical characteristics. This resulted in overlapping prediction patterns and relatively low R^2 values, suggesting that physicochemical measurements alone may not fully capture human sensory perception of wine quality. From the experiments, the group gained a better understanding of how preprocessing, cross-validation, feature scaling, residual analysis, and confusion matrix evaluation contribute to more reliable machine learning workflows.

Future Improvement

Several limitations were identified during the project, including dataset size and class imbalance, which may have affected model generalisation performance. Future work could explore ordinal encoding and ordinal classification which can preserve the ordering of the classes compared to ordinary classification. Class imbalance could also be addressed using resampling techniques such as SMOTE or threshold adjustment. In addition, nested cross-validation could provide a more reliable estimate of tuned model performance, while external validation datasets could further evaluate model generalisation beyond the current dataset. Stronger regression models like Support vector regression or XGBoost could also be used instead.

5. AcKnowledge

FT14					
Section	Xuexin Deng	Jin Wan Kim	Ivan Koh Zhi En	Renyu Xia	Yuxuan Yan
1. Introduction	S 20%		L 80%		
2. Background Theory					
2.1 Identify the Quality of Wine			L 80%		S 20%
2.2 Data Cleaning, Preprocessing and Analysis		S 20%		L 80%	
2.3 Regression Models		L 80%	S 20%		
2.4 Classification Models	S 20%				L 80%
2.5 Model Evaluation Metrics and Method	L 50%			S 50%	
3. Solutions, Evaluation, and Discussions					
3.1 Data Cleaning and Preprocessing	S 30%		L 70%		
3.2 Regression Model Setup and Fine-tune		L 70%		S 30%	
3.3 Regression Model Evaluation		S 30%		L 70%	
3.4 Classification Model Setup, Comparison and Fine-tune			S 30%		L 70%
3.5 Random Forest Classifier Model Evaluation	L 70%				S 30%
4. Conclusion & References	S 50%		L 50%		

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